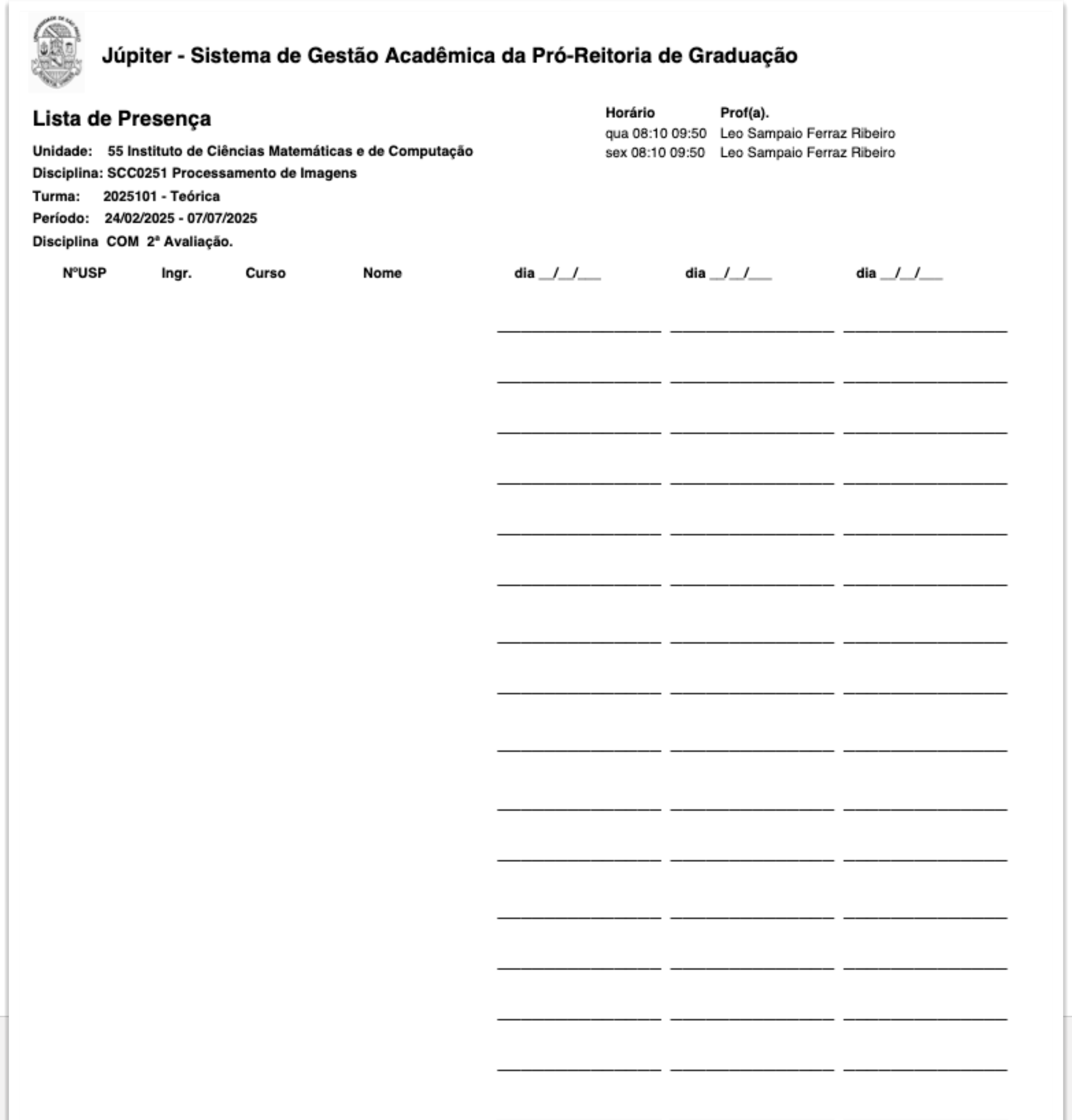
Processamento de Imagens

Filtros e Convolução (Parte II)

Professora Leo Sampaio Ferraz Ribeiro

Slide para não esquecer de passar a lista



Júpiter - Sistema de Gestão Acadêmica da Pró-Reitoria de Graduação

Lista de Presença

Unidade: 55 Instituto de Ciências Matemáticas e de Computação
Disciplina: SCC0251 Processamento de Imagens
Turma: 2025101 - Teórica
Período: 24/02/2025 - 07/07/2025
Disciplina COM 2ª Avaliação.

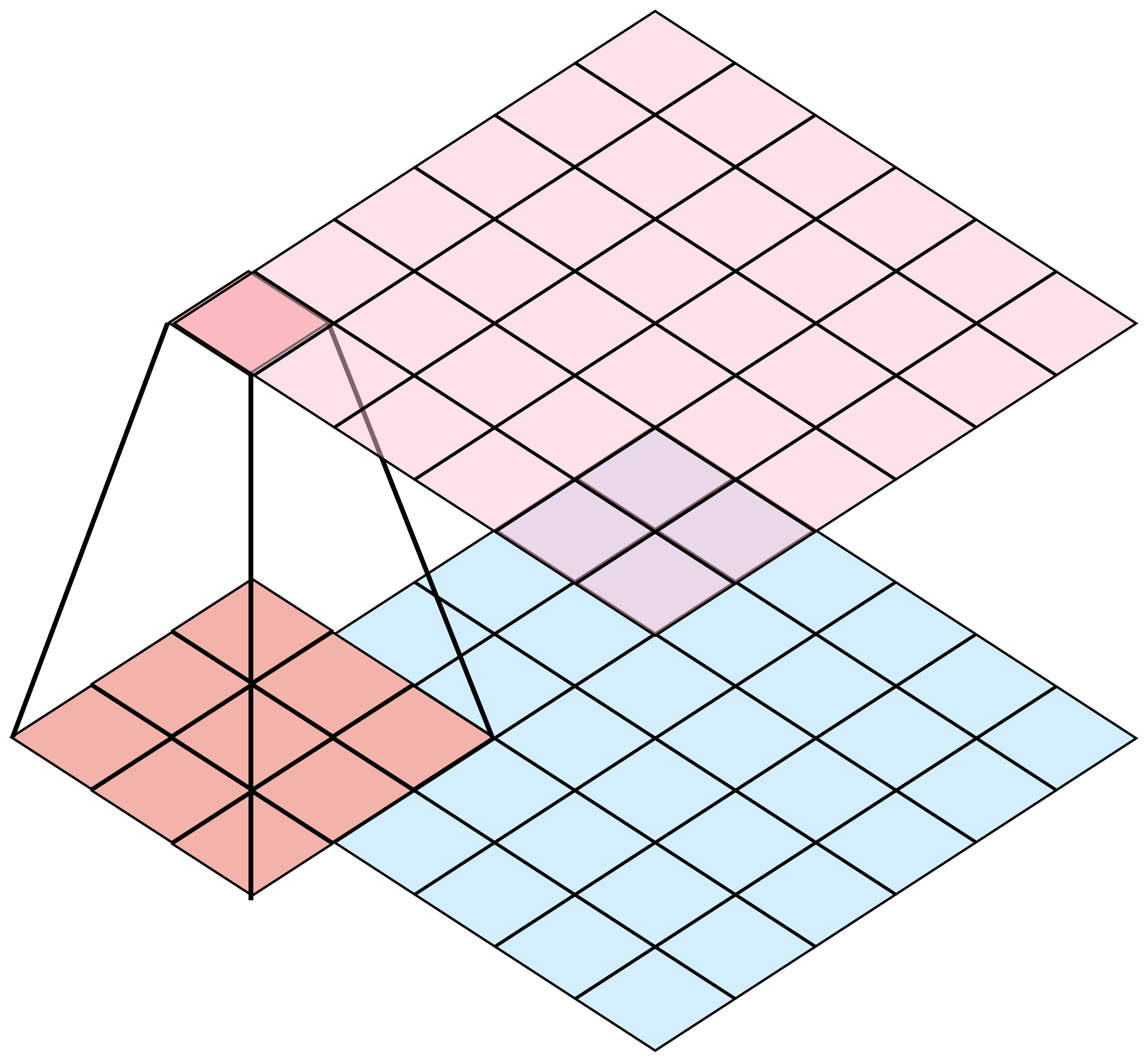
Horário	Prof(a).
qua 08:10 09:50	Leo Sampaio Ferraz Ribeiro
sex 08:10 09:50	Leo Sampaio Ferraz Ribeiro

NºUSP	Ingr.	Curso	Nome	dia __/__/___	dia __/__/___	dia __/__/___

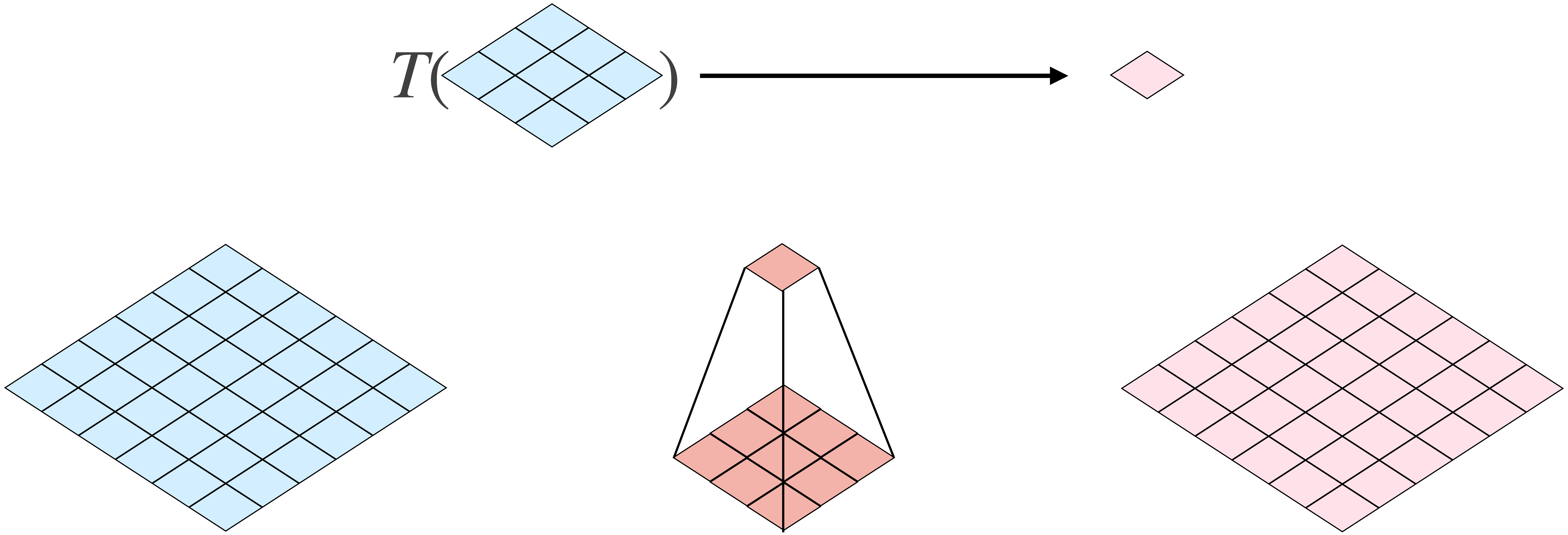
Ar Condicionado está ok?

Dúvidas de Trabalhos e Exercícios?

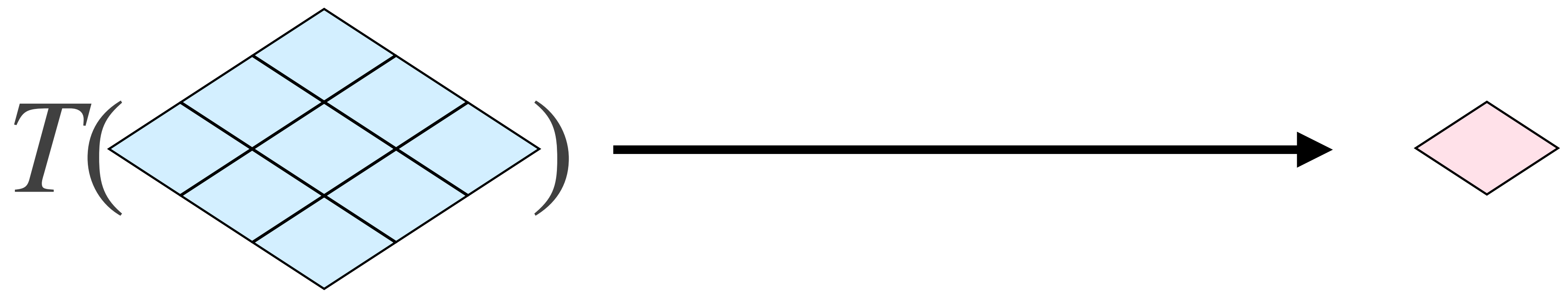
Convolução

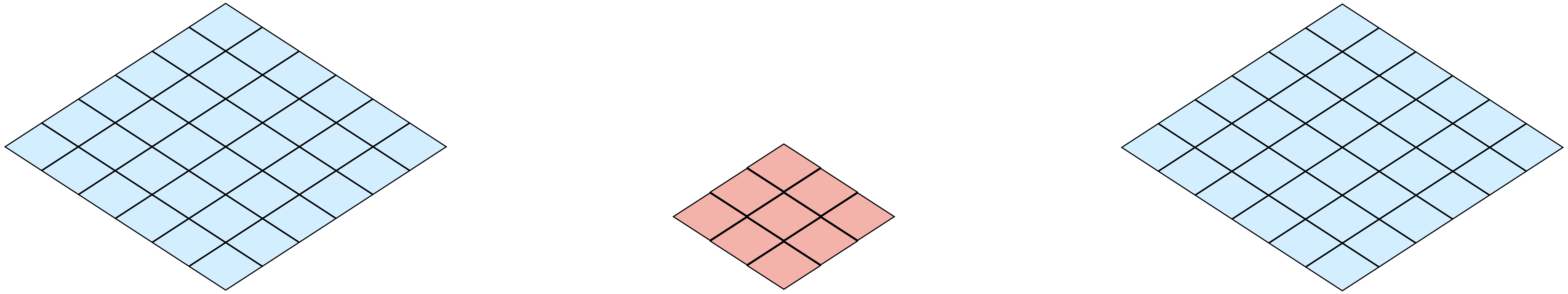


Tipos de Filtros



Filtros de Shift





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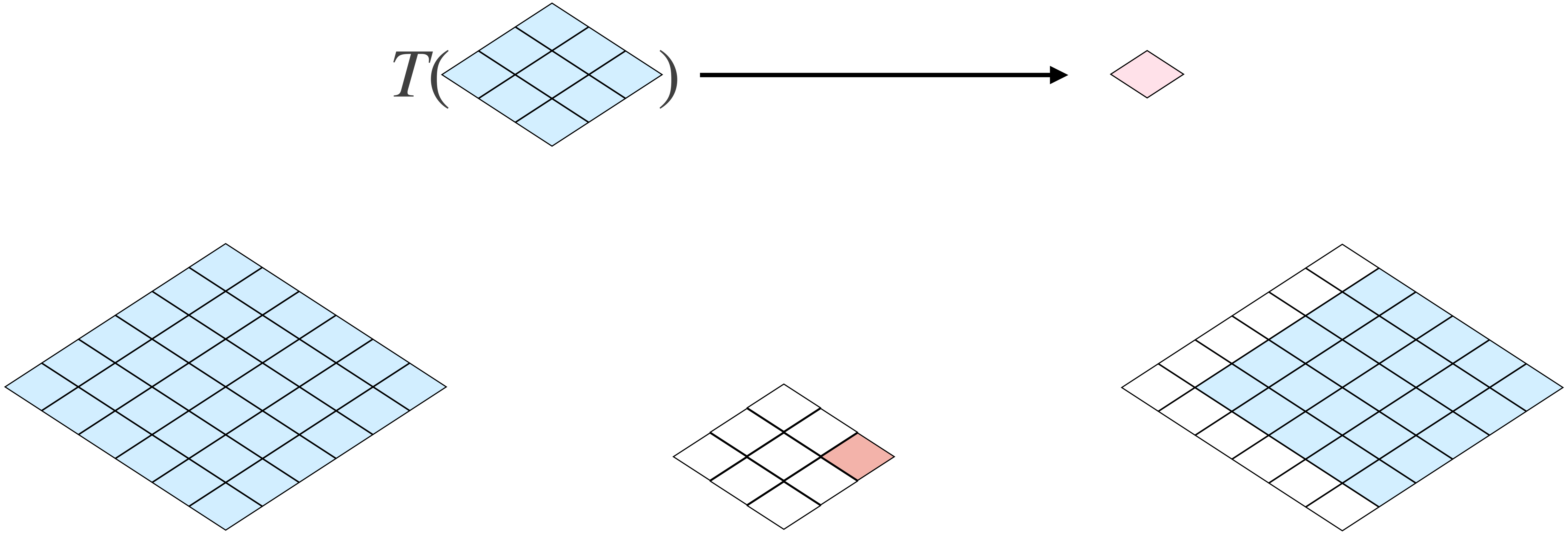
Filtros de Shift



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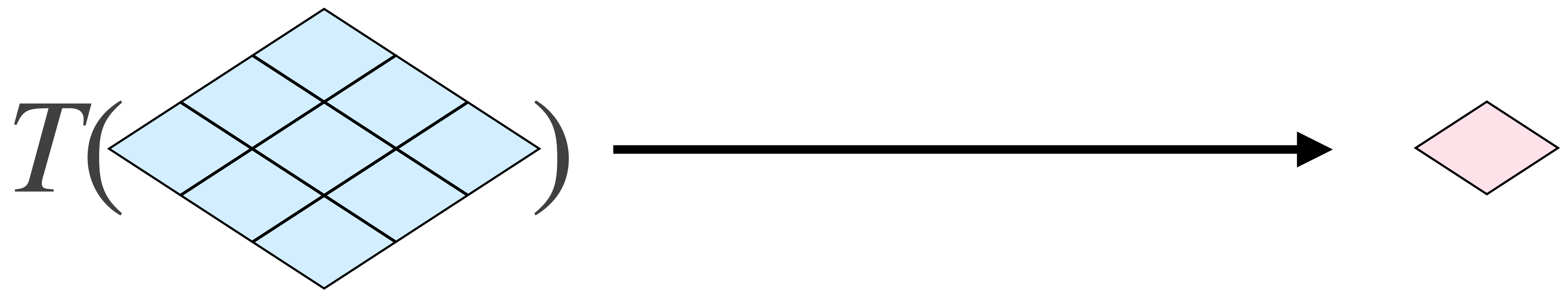
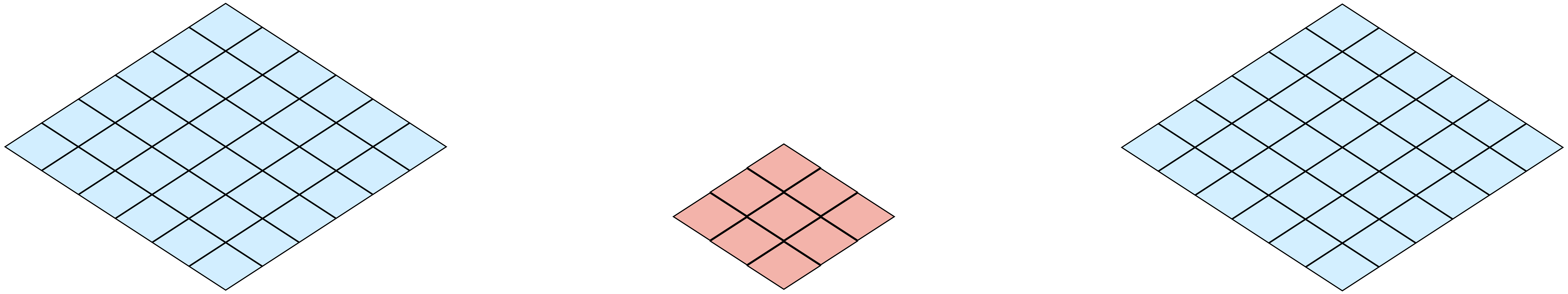
Filtros de Shift

```
$ python demo.py
```

Filtros de Caixa





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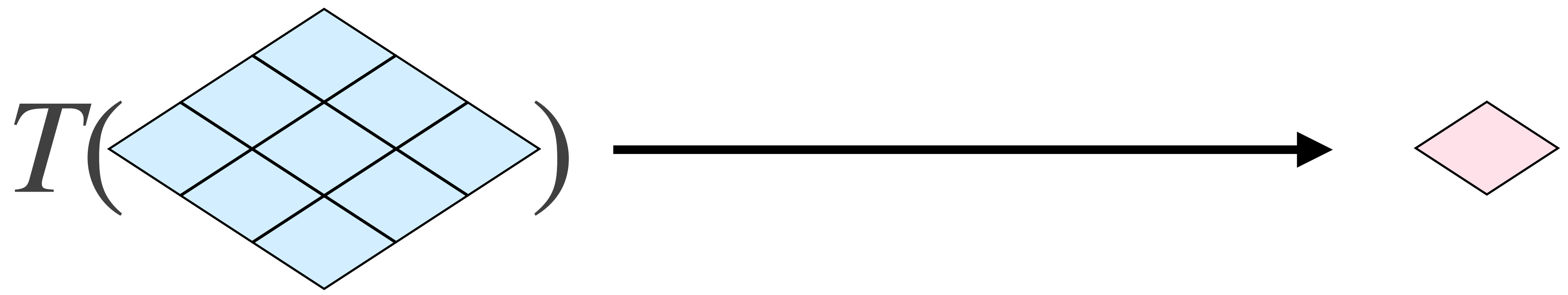
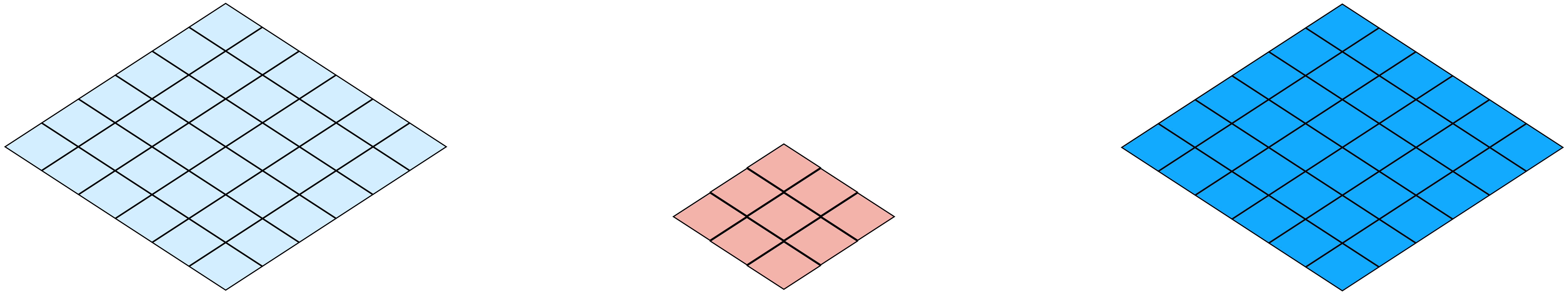
1	1	1

Filtros de Caixa





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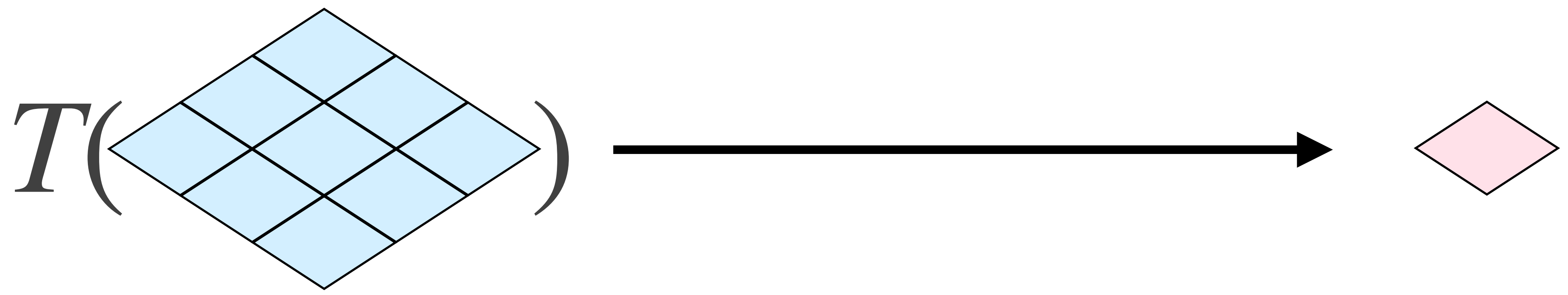
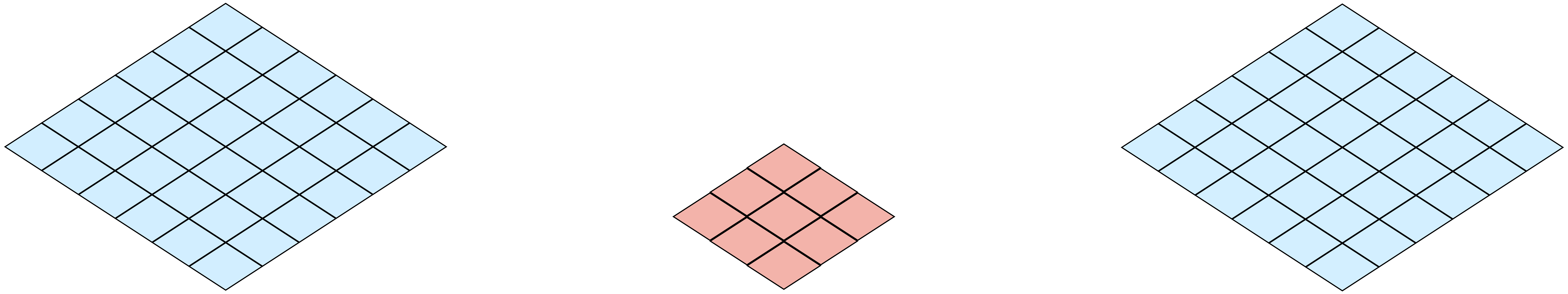
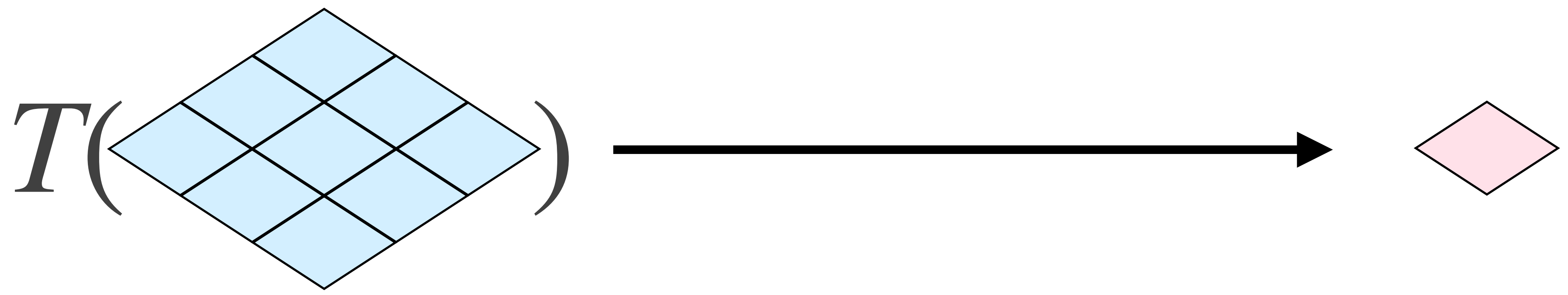
Filtros de Caixa

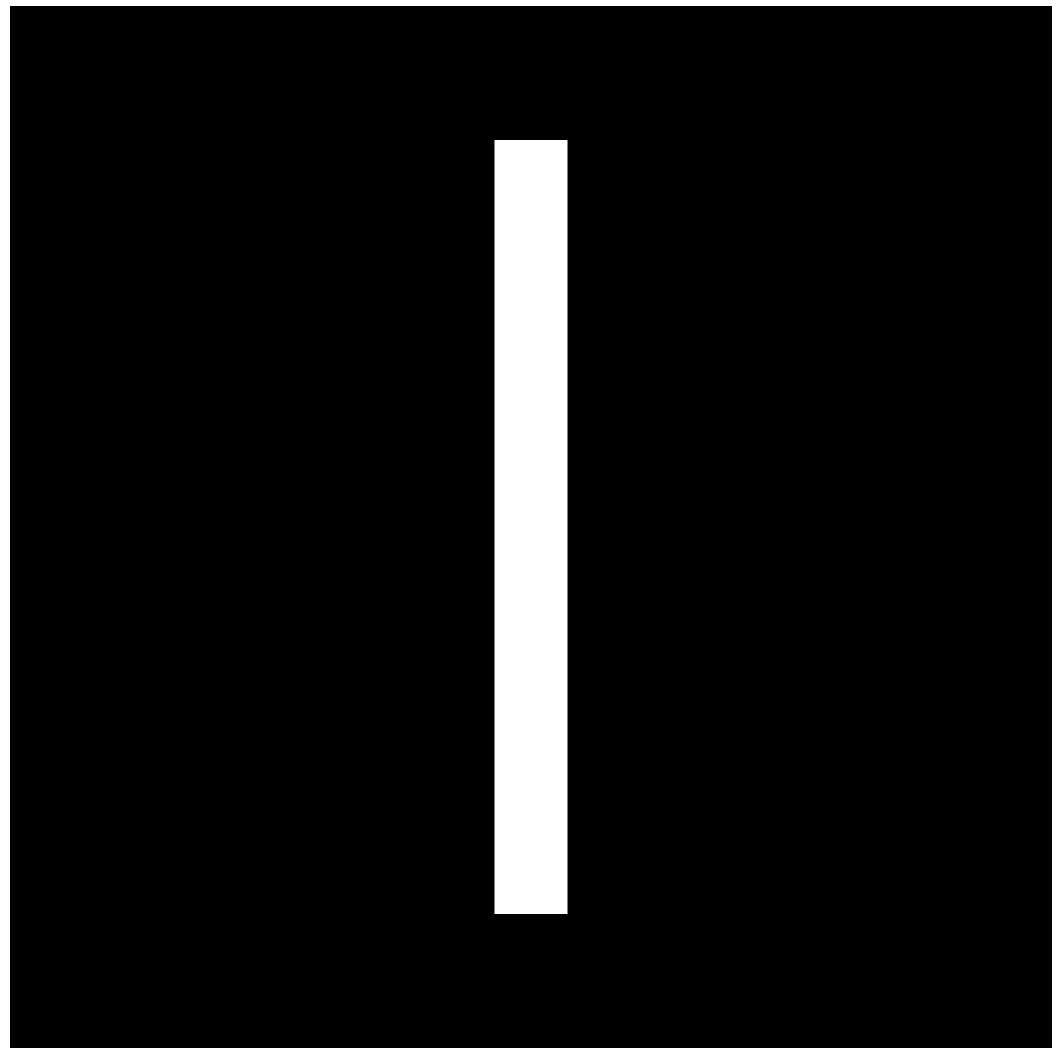


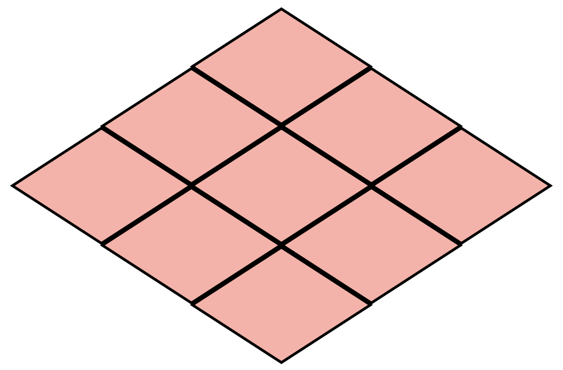


1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

Filtros de Caixa







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1/9	1/9	1/9
1/9	1/9	1/9

Filtros de Caixa

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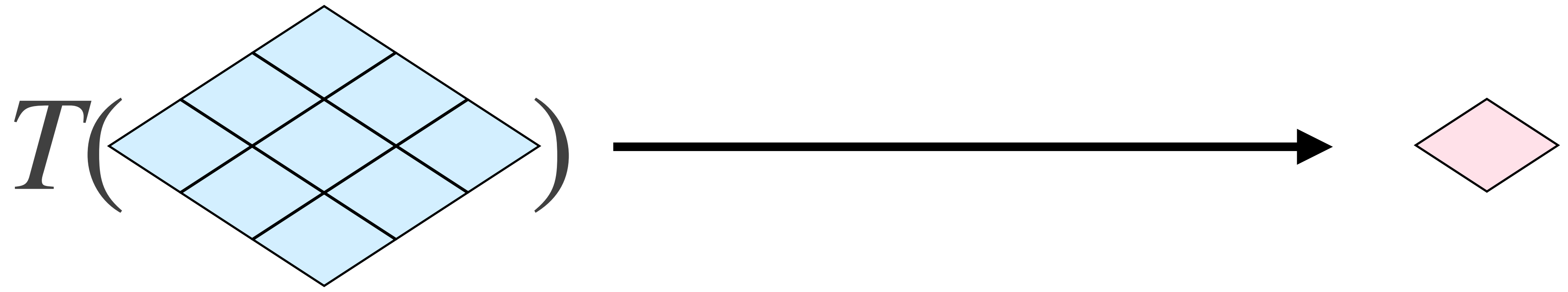
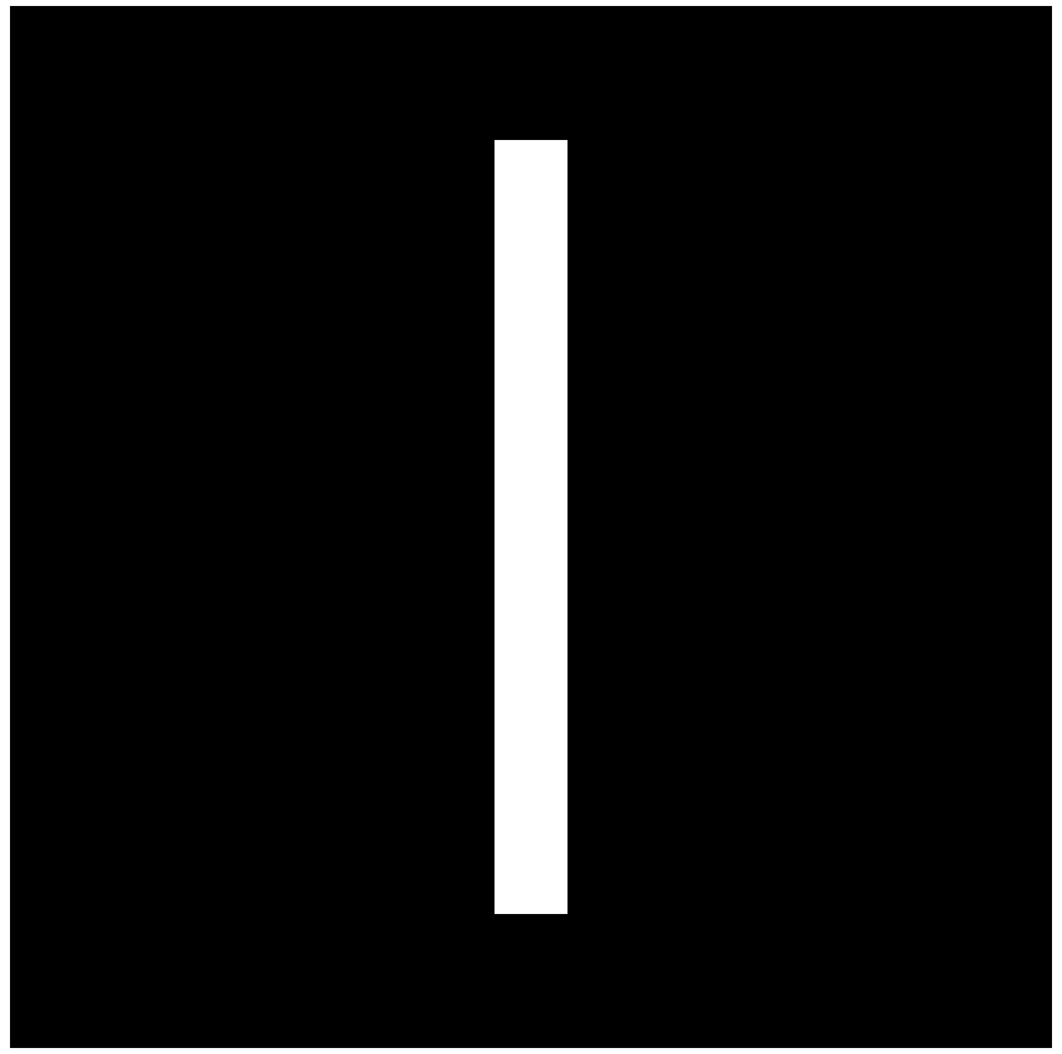
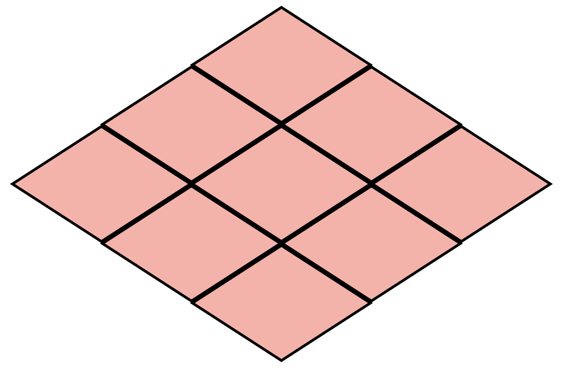
$ python demo.py
```

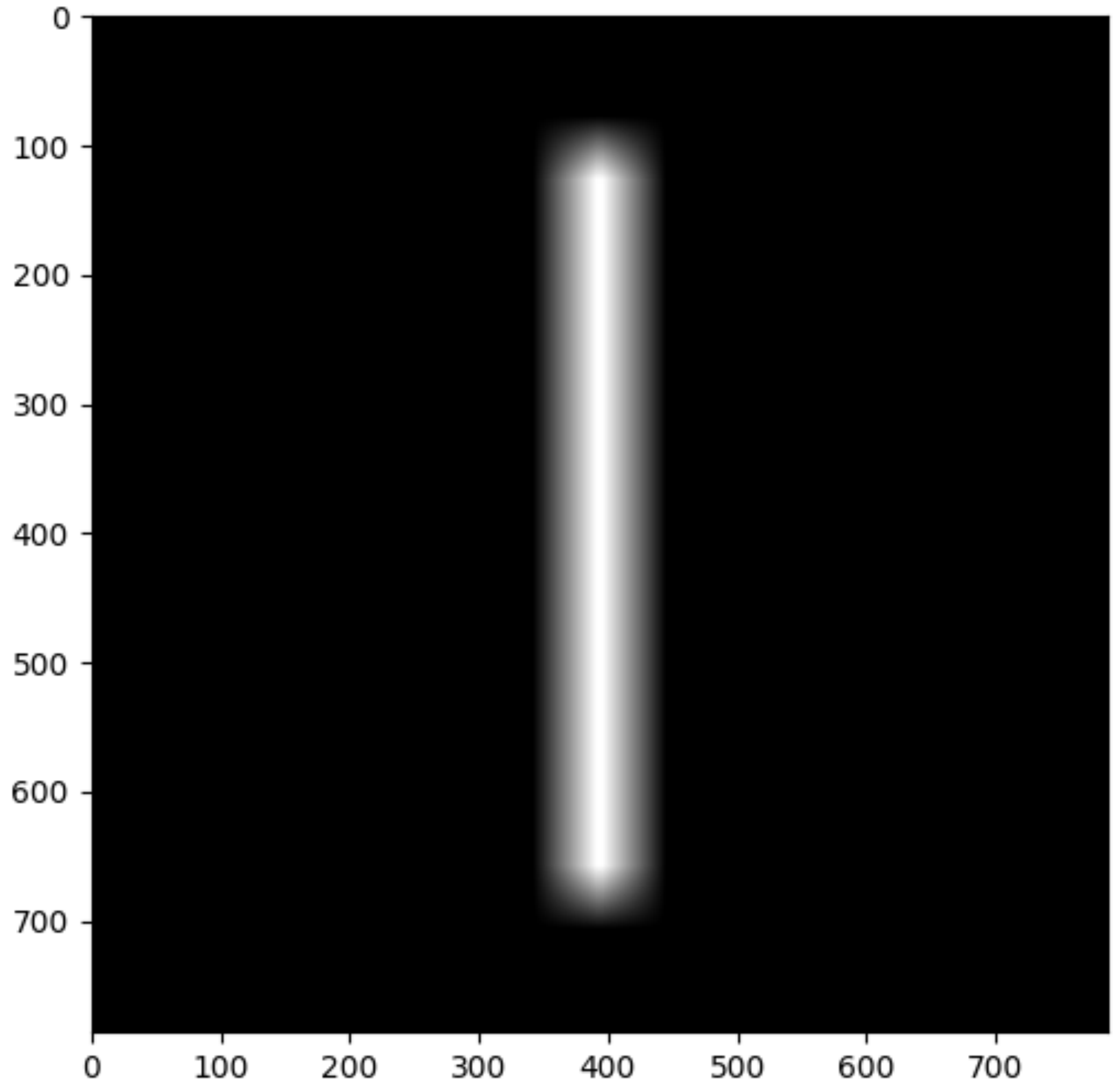
Filtros de Caixa



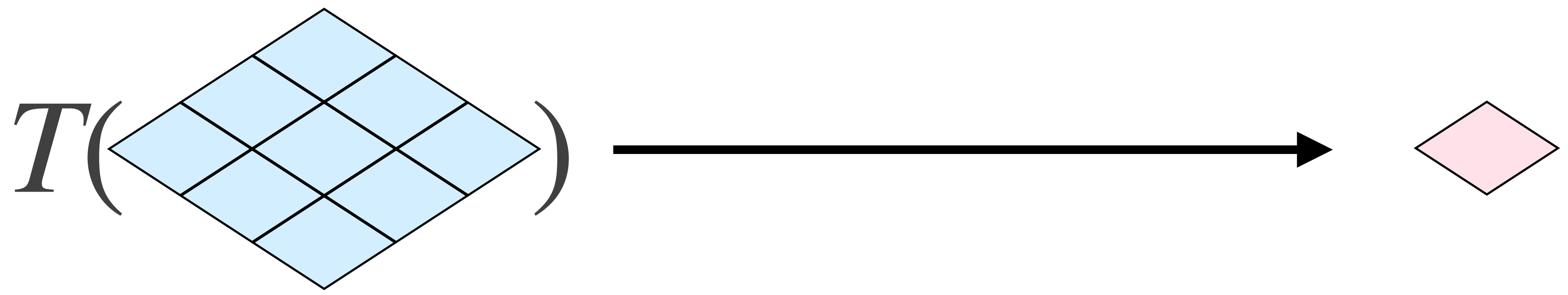


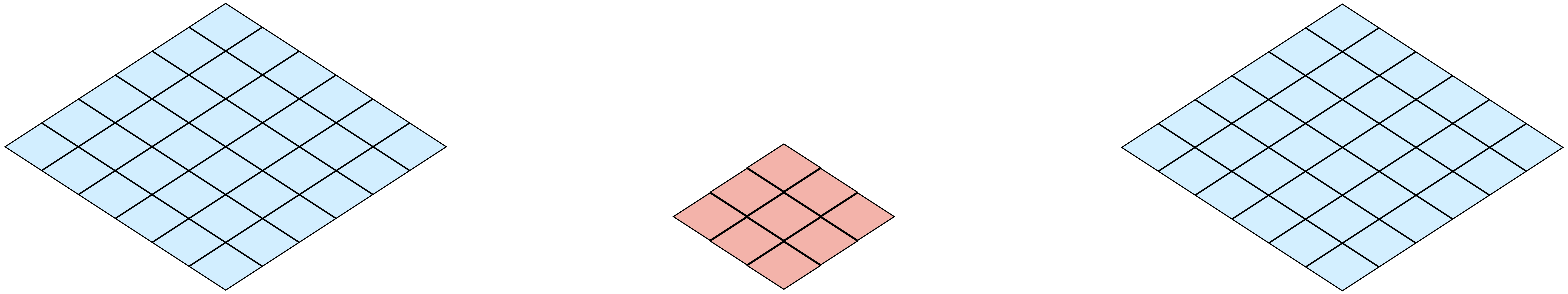


1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9



Filtros Gaussianos





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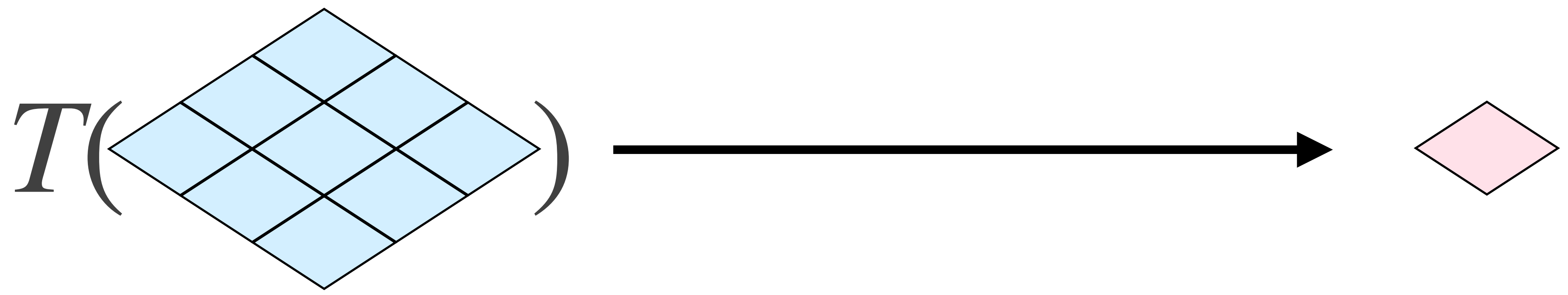
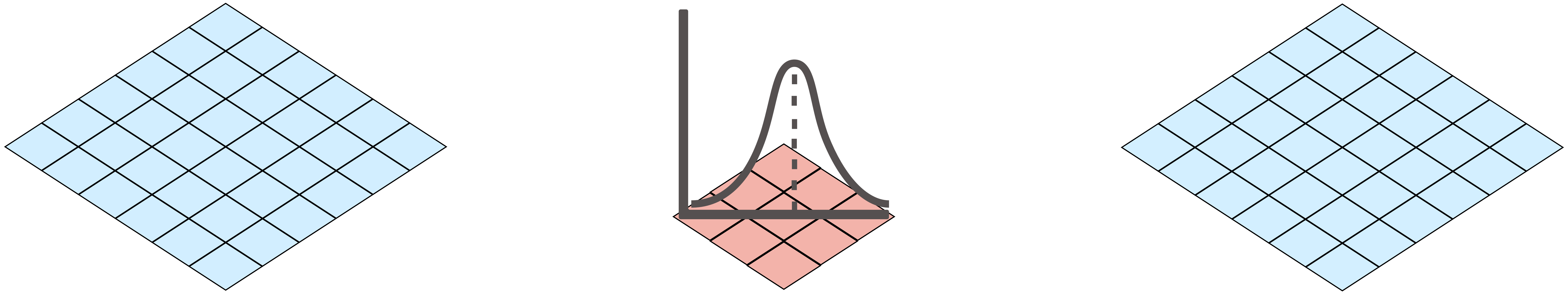
Filtros Gaussianos





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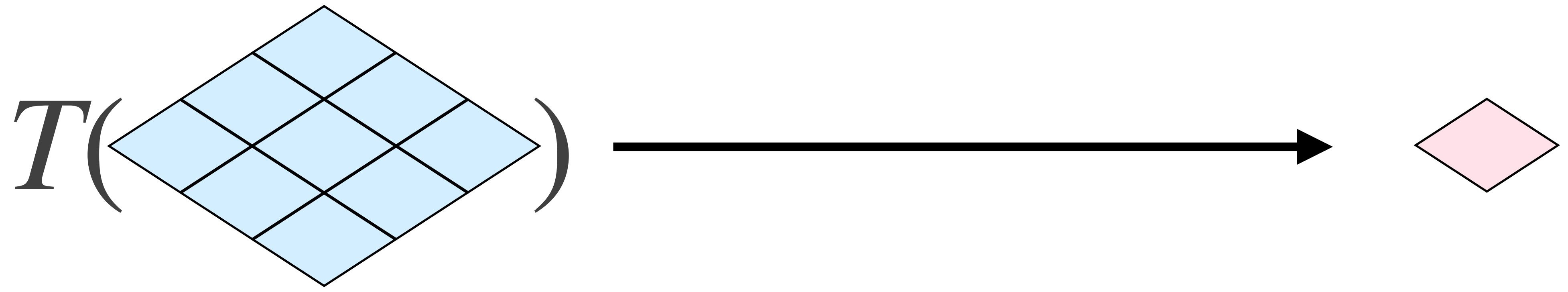
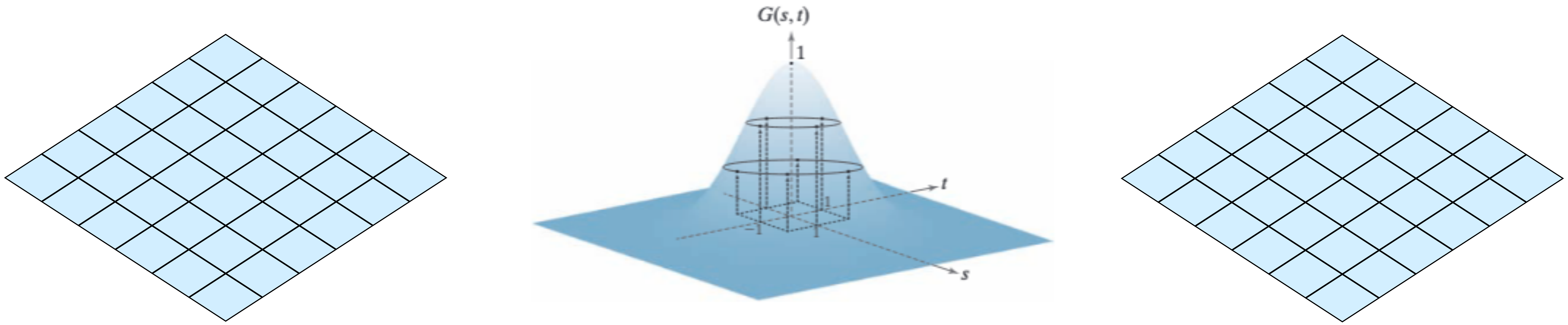
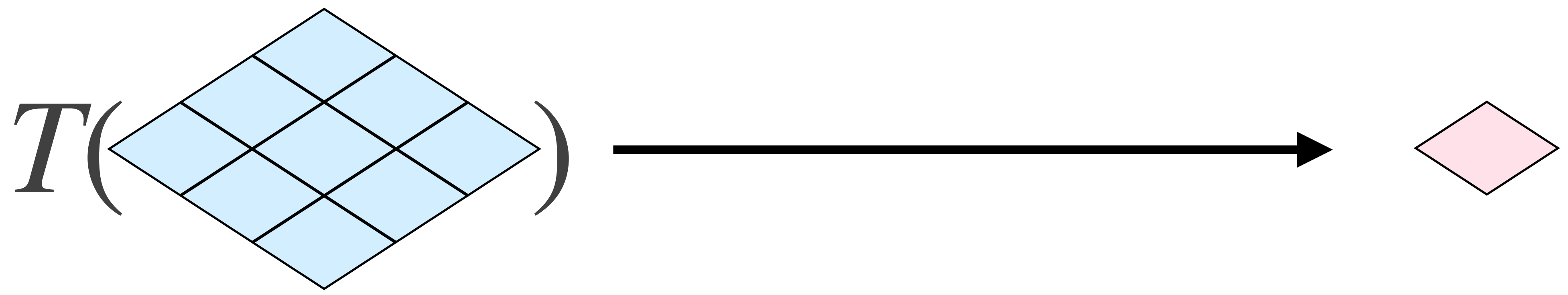
Filtros Gaussianos

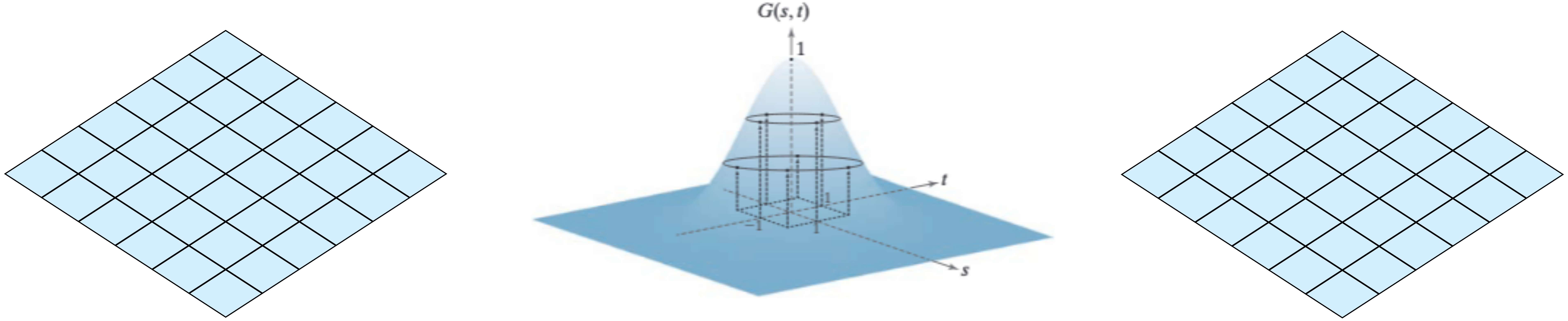




$$\frac{1}{4.8976} \times \begin{array}{|c|c|c|}\hline 0.3679 & 0.6065 & 0.3679 \\ \hline 0.6065 & 1.0000 & 0.6065 \\ \hline 0.3679 & 0.6065 & 0.3679 \\ \hline \end{array}$$

Filtros Gaussianos



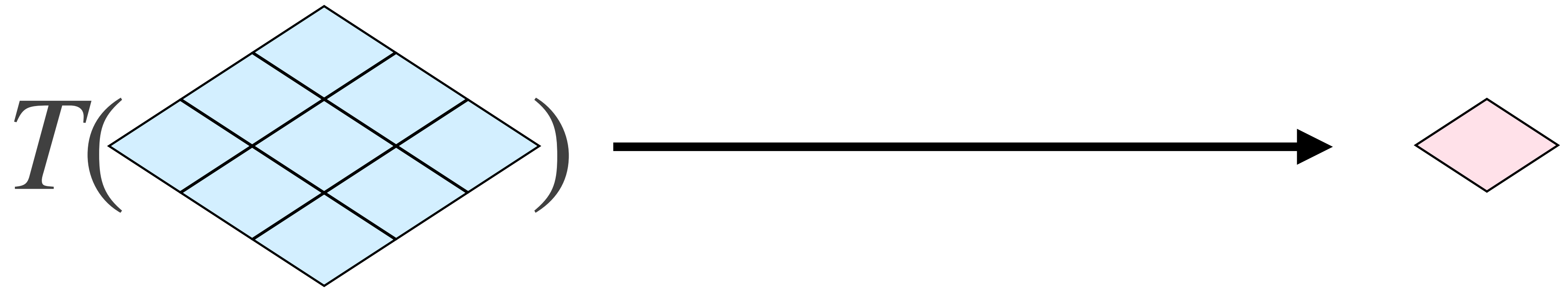


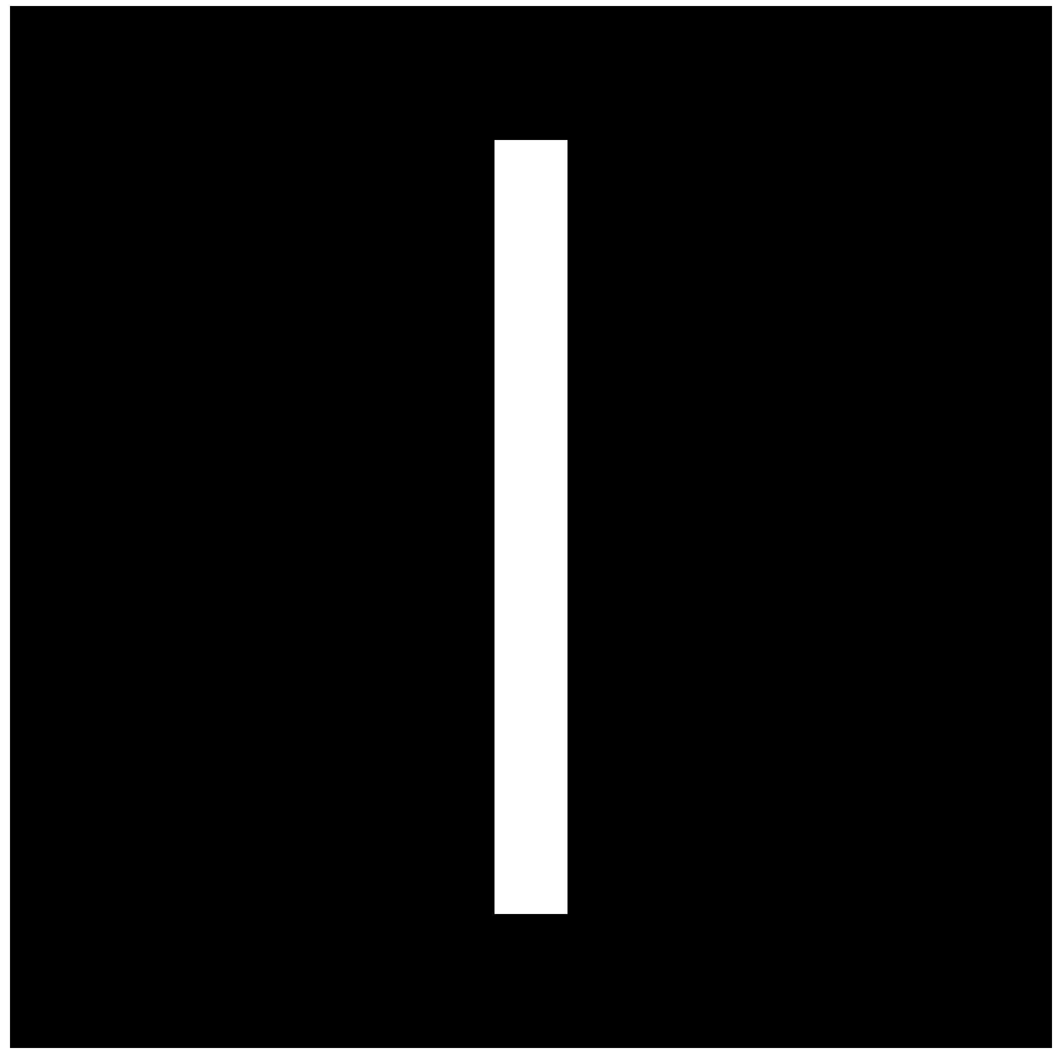
$$G_{1D}(x,\sigma) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{x^2}{2\sigma^2}} \qquad G_{2D}(x,y,\sigma) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

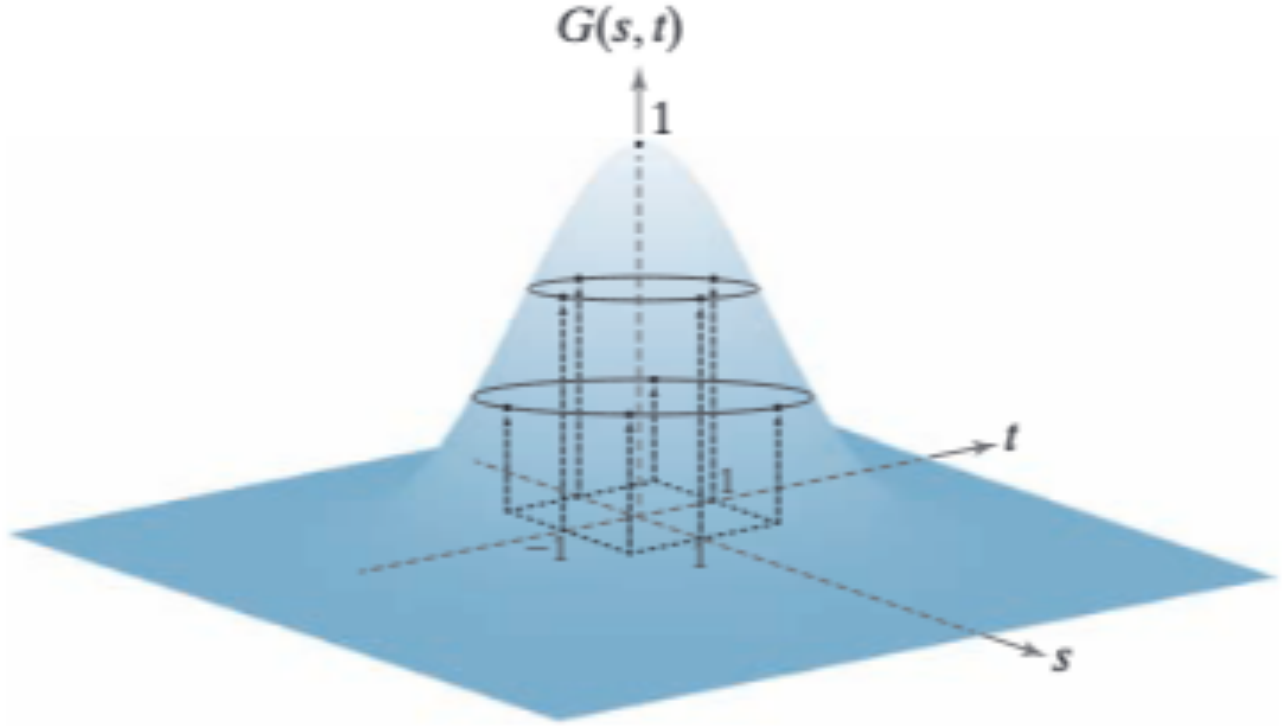
Filtros Gaussianos

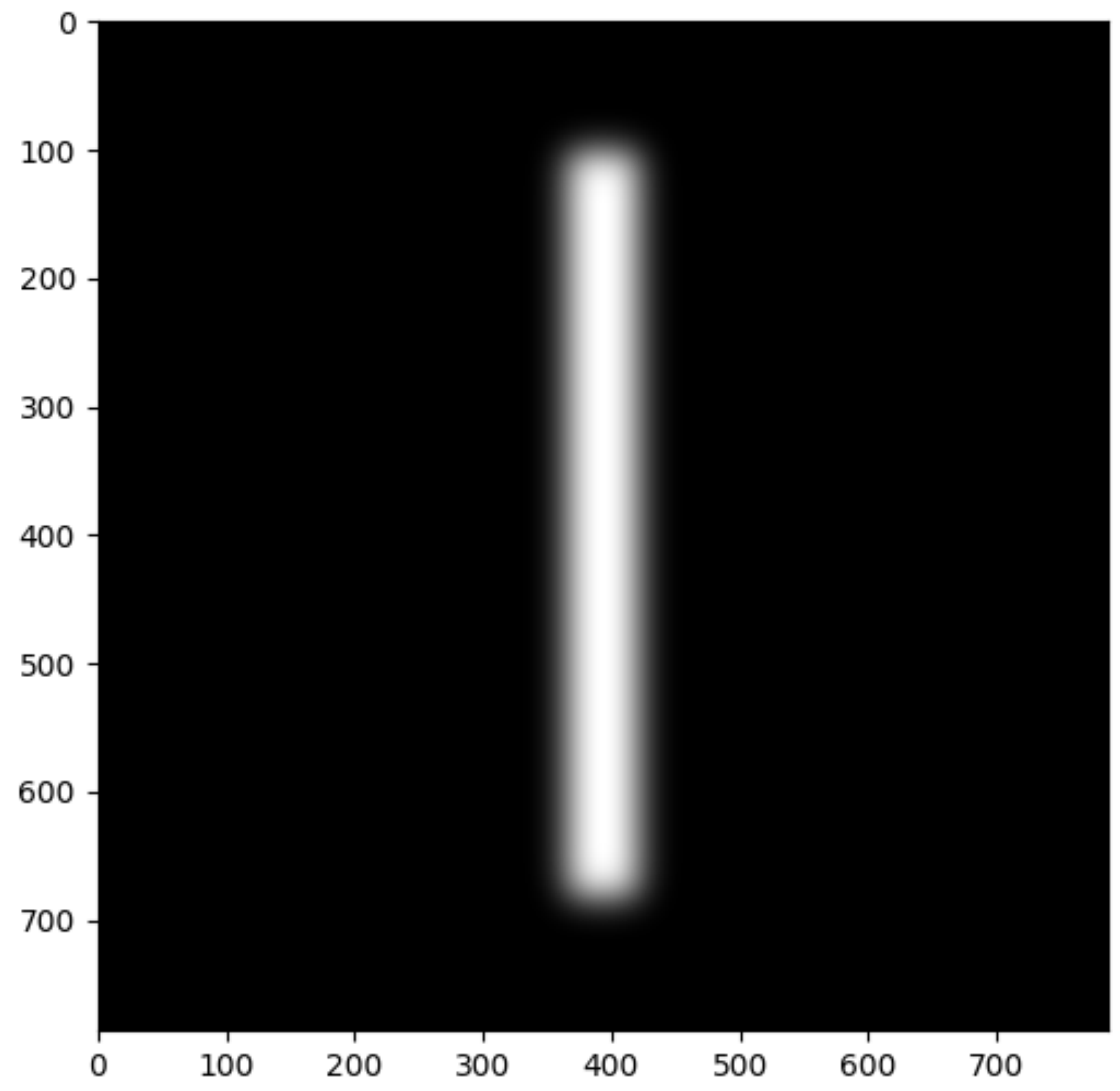
```
$ python demo.py
```

Filtros de Caixa



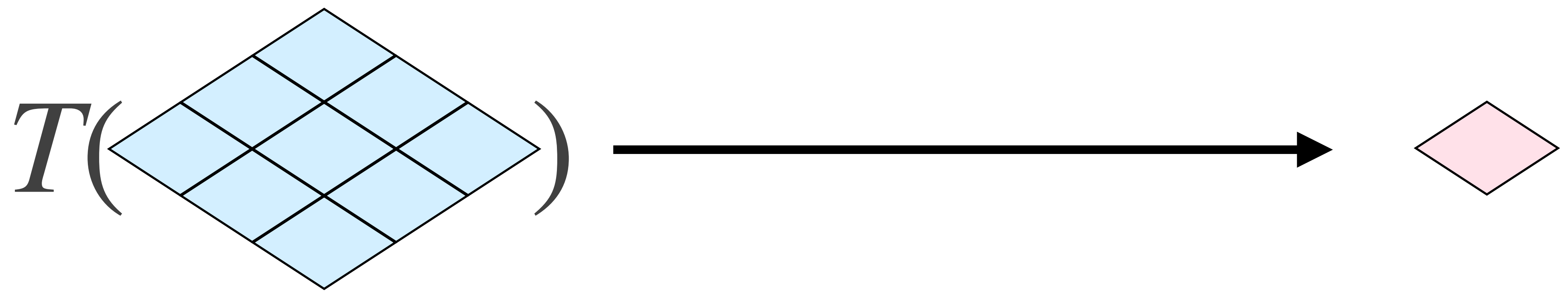


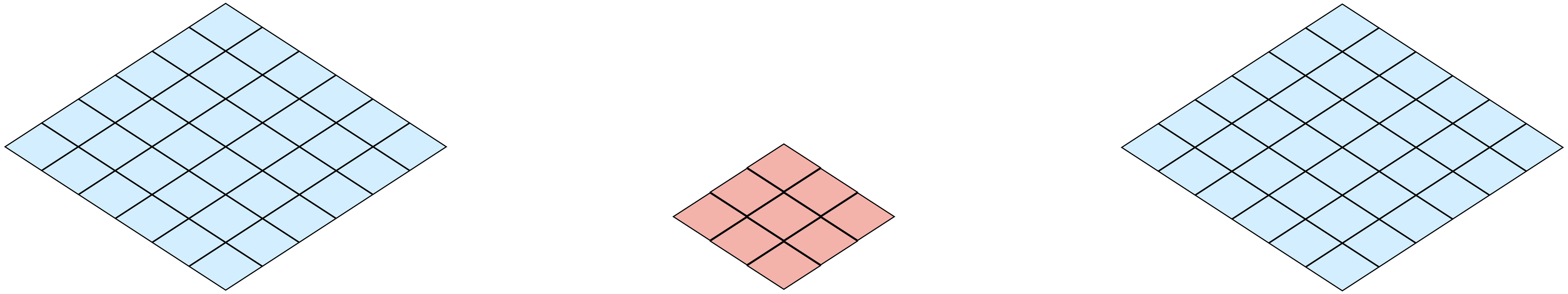




$$G_{2D}(x, y, \sigma) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2 + y^2}{2\sigma^2}}$$

Filtros de Mediana





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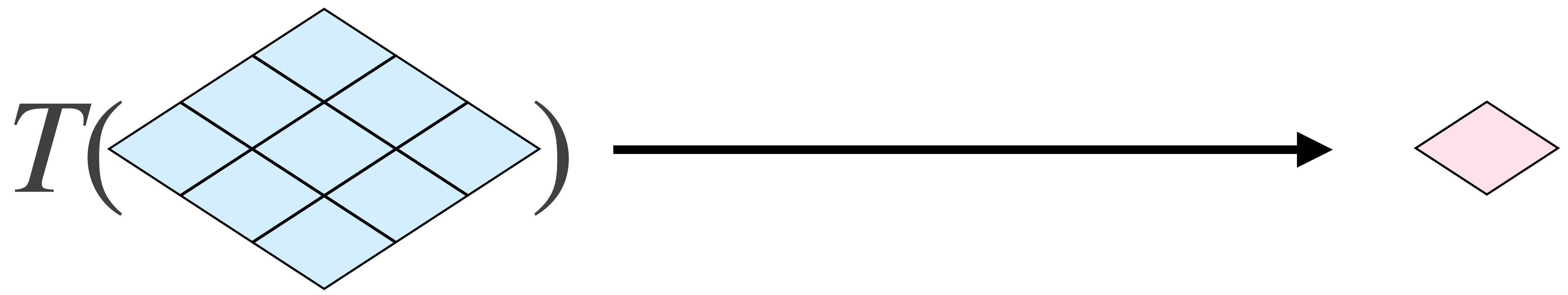
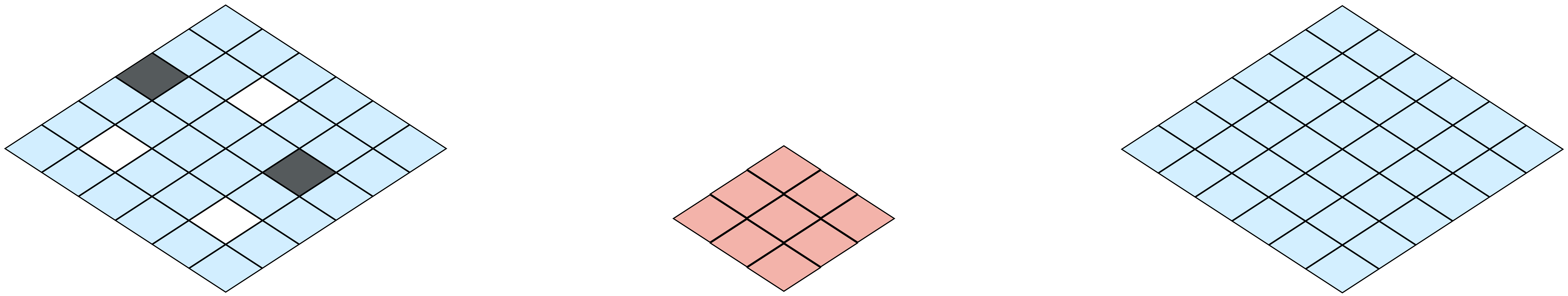
Filtros de Mediana





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Filtros de Mediana

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$ python demo.py
```
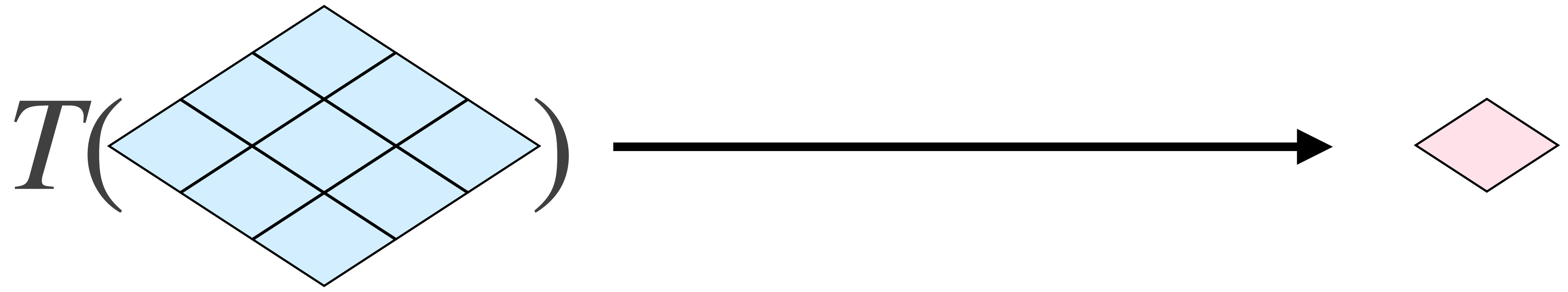
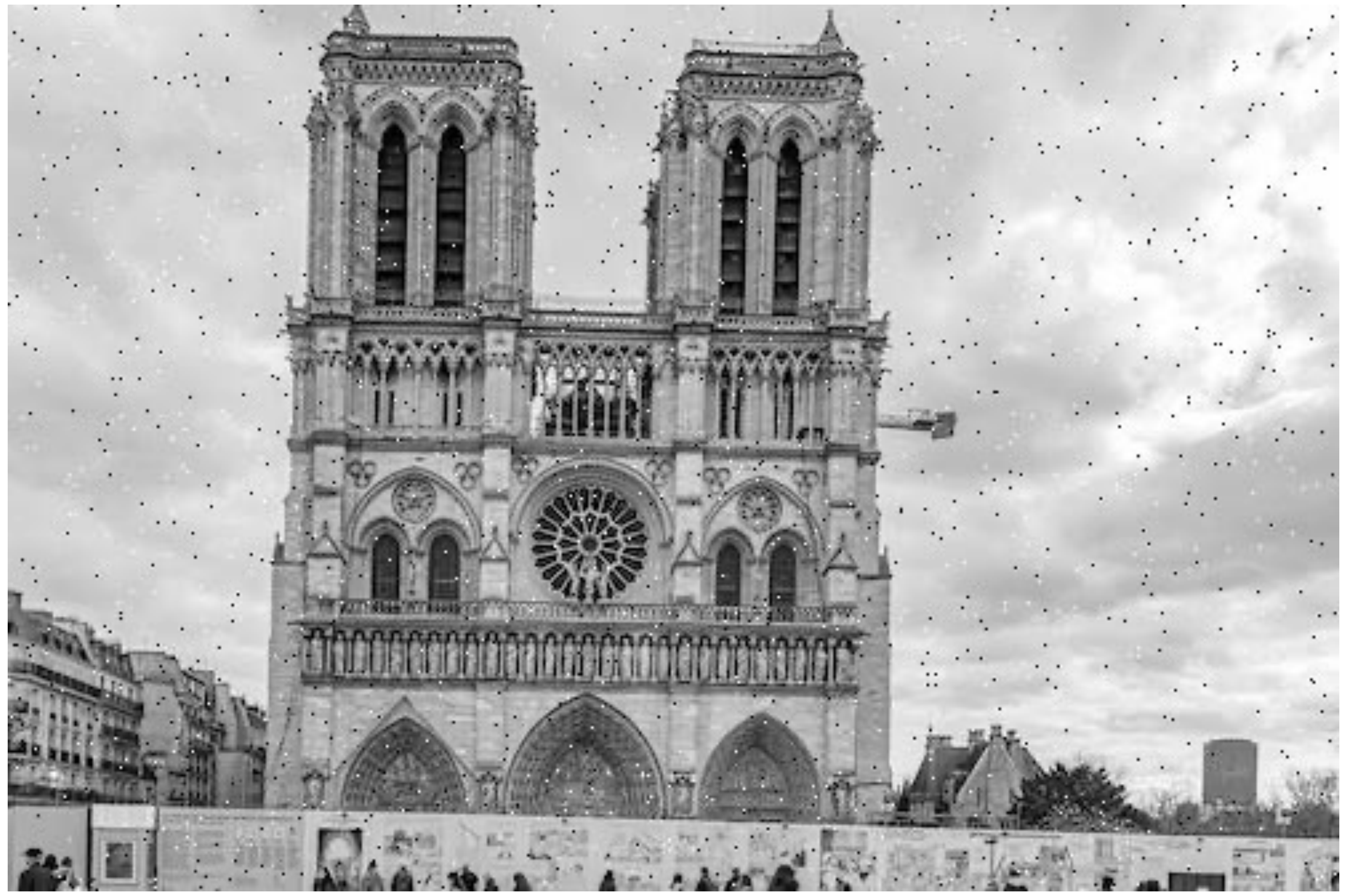
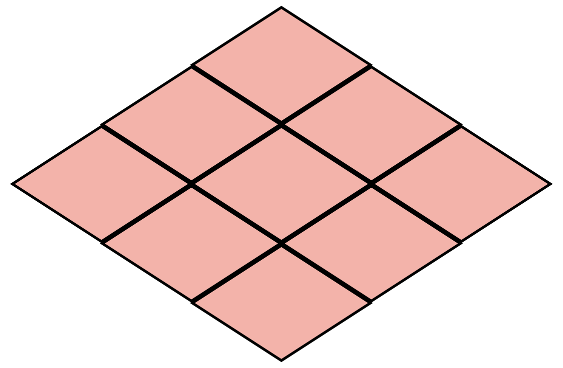

Filtros de Mediana







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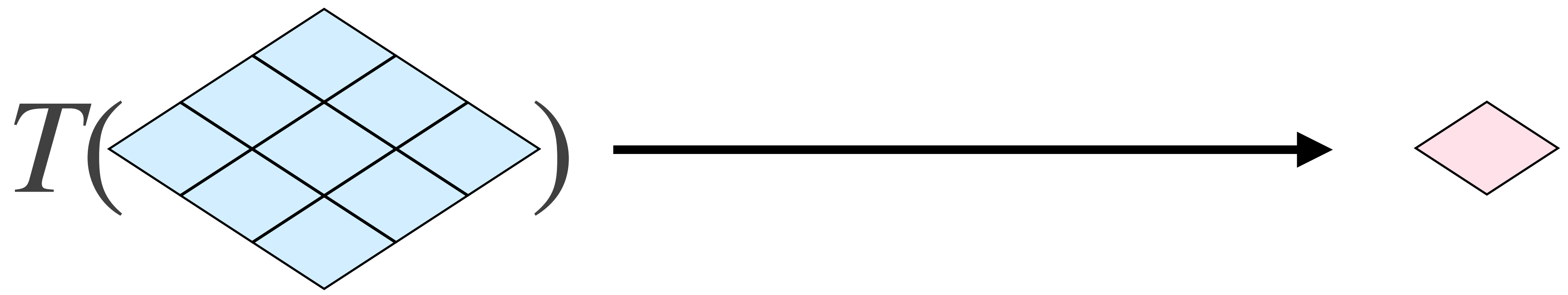
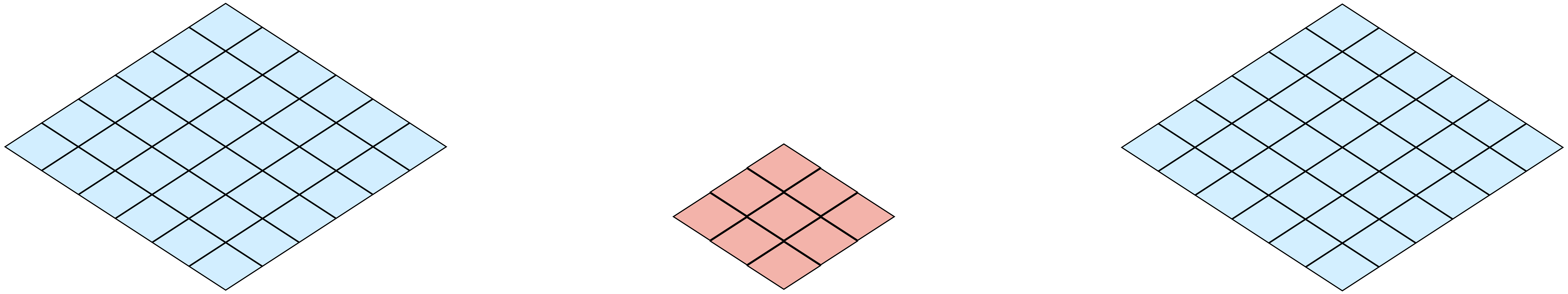
Filtros de Sharpening





?	?	?
?	?	?
?	?	?

Derivadas Discretas

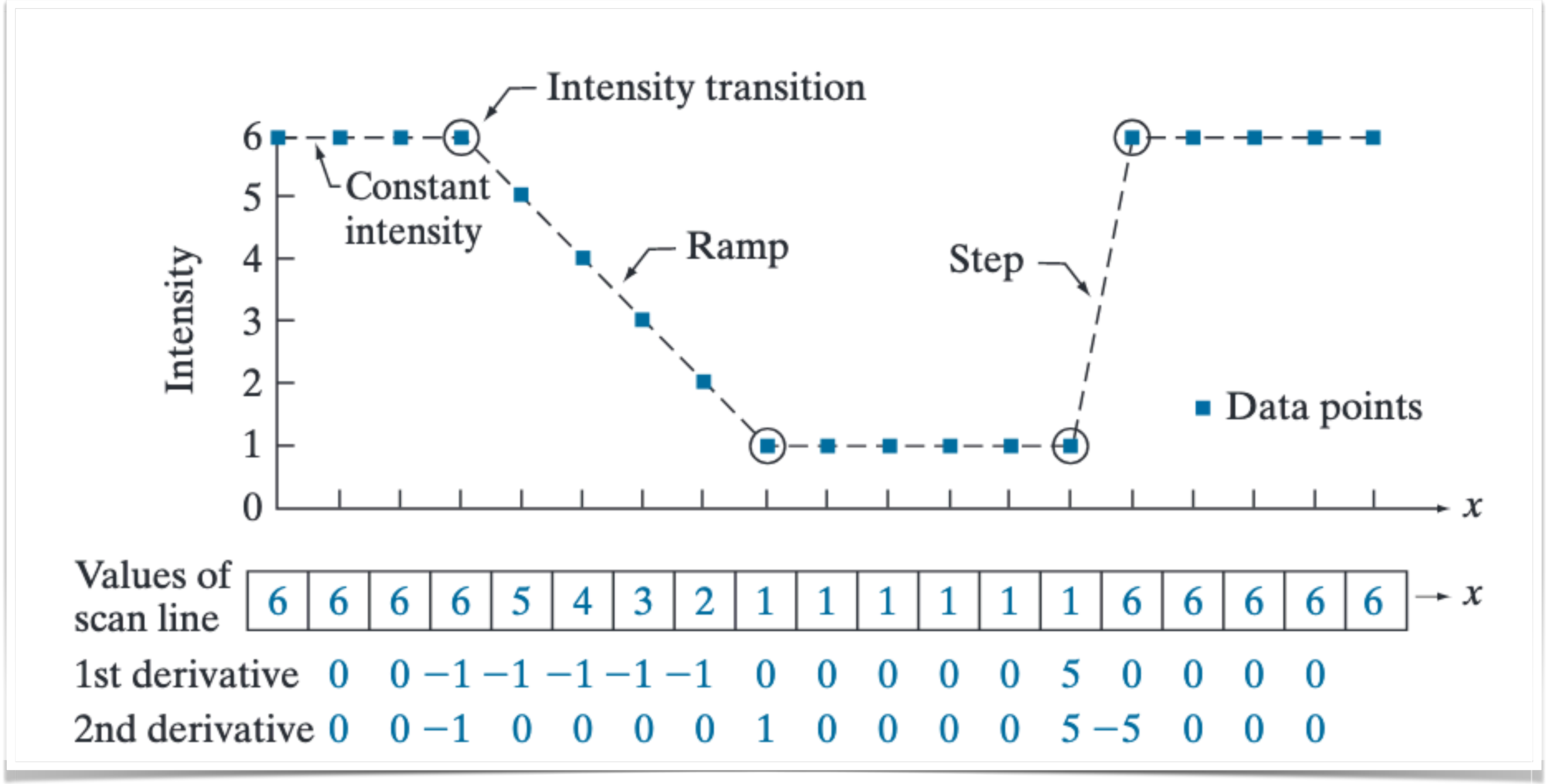
Derivada de 1ª Ordem

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

Derivada de 2ª Ordem

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$

Derivadas Discretas



Derivadas Discretas

Operador Isotrópico: Laplace

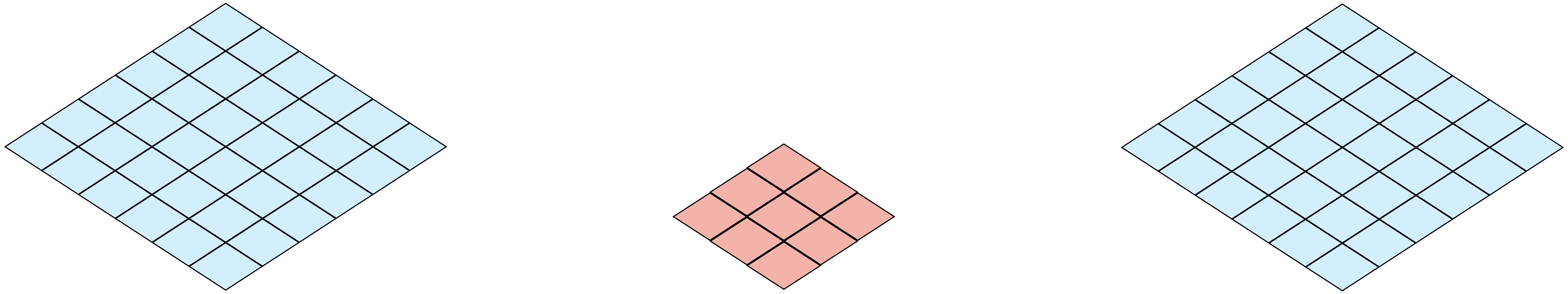
$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

Aproximação discreta:

$$\nabla^2 f = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

Filtros de Sharpening

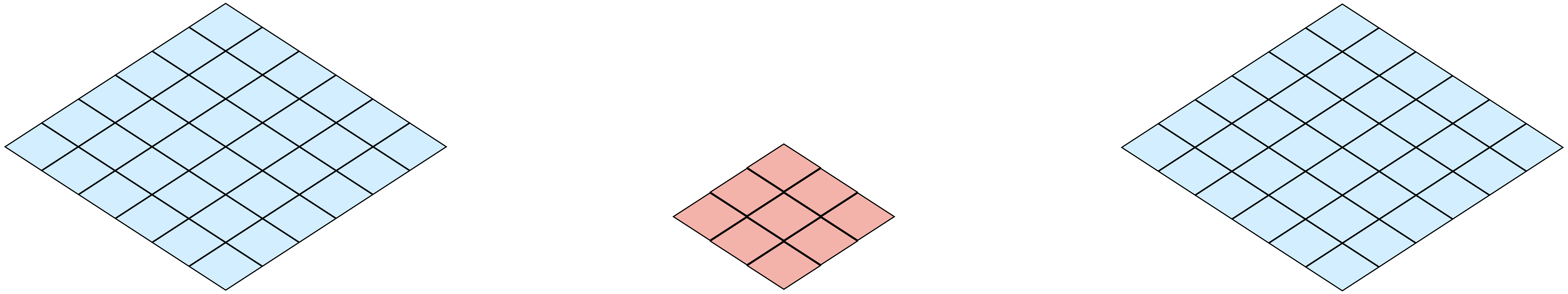
$$\nabla^2 f = f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1) - 4f(x,y)$$



?	?	?
?	?	?
?	?	?

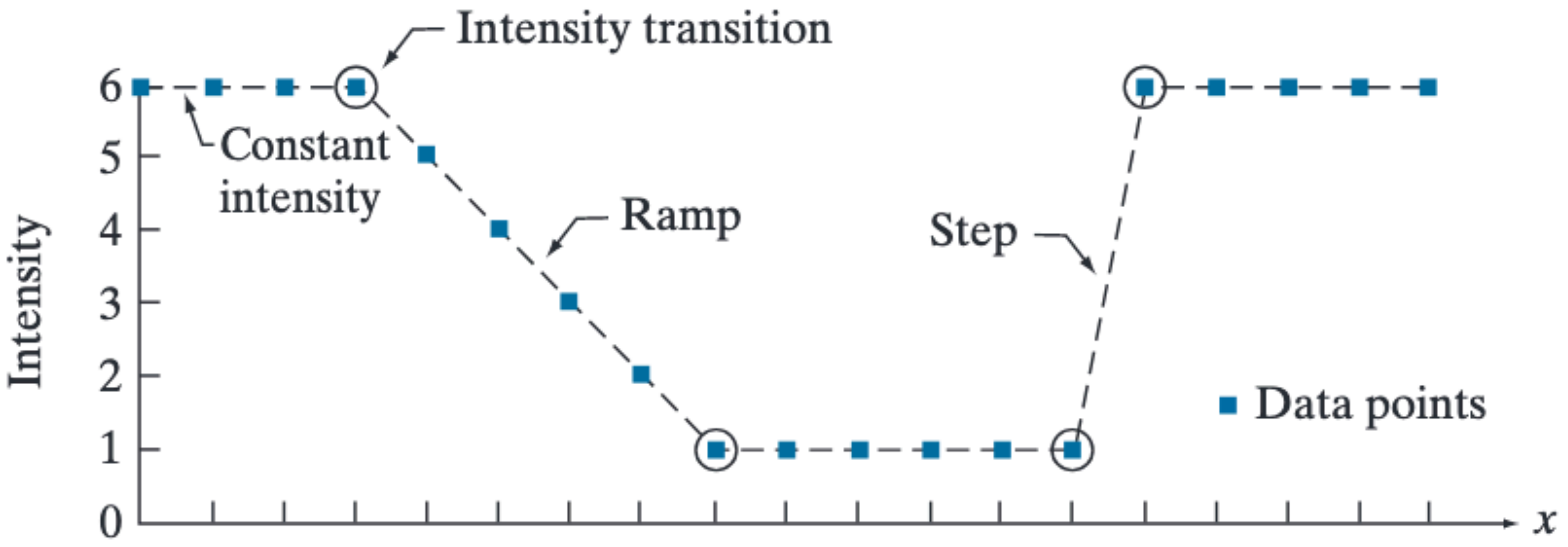
Filtros de Sharpening

$$\nabla^2 f = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$



0	1	0
1	-4	1
0	1	0

Filtros de Sharpening

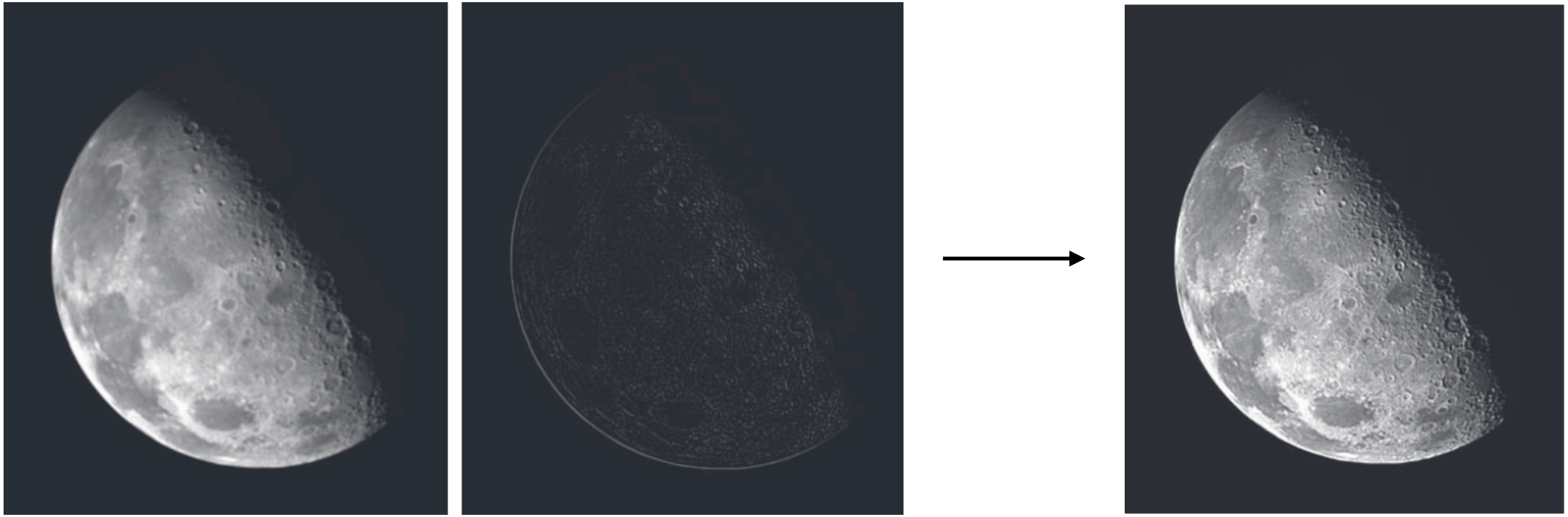


Values of scan line	6	6	6	6	5	4	3	2	1	1	1	1	1	1	6	6	6	6	6
1st derivative		0	0	−1	−1	−1	−1	−1	0	0	0	0	0	5	0	0	0	0	
2nd derivative		0	0	−1	0	0	0	0	1	0	0	0	0	5	−5	0	0	0	

Filtros de Sharpening

Desejamos então **adicionar as derivadas de segunda ordem na imagem**, aumentando a diferença nas bordas

$$g(x, y) = f(x, y) + c\,|\,\nabla^2 f(x, y)\,|$$



Filtros de Sharpening

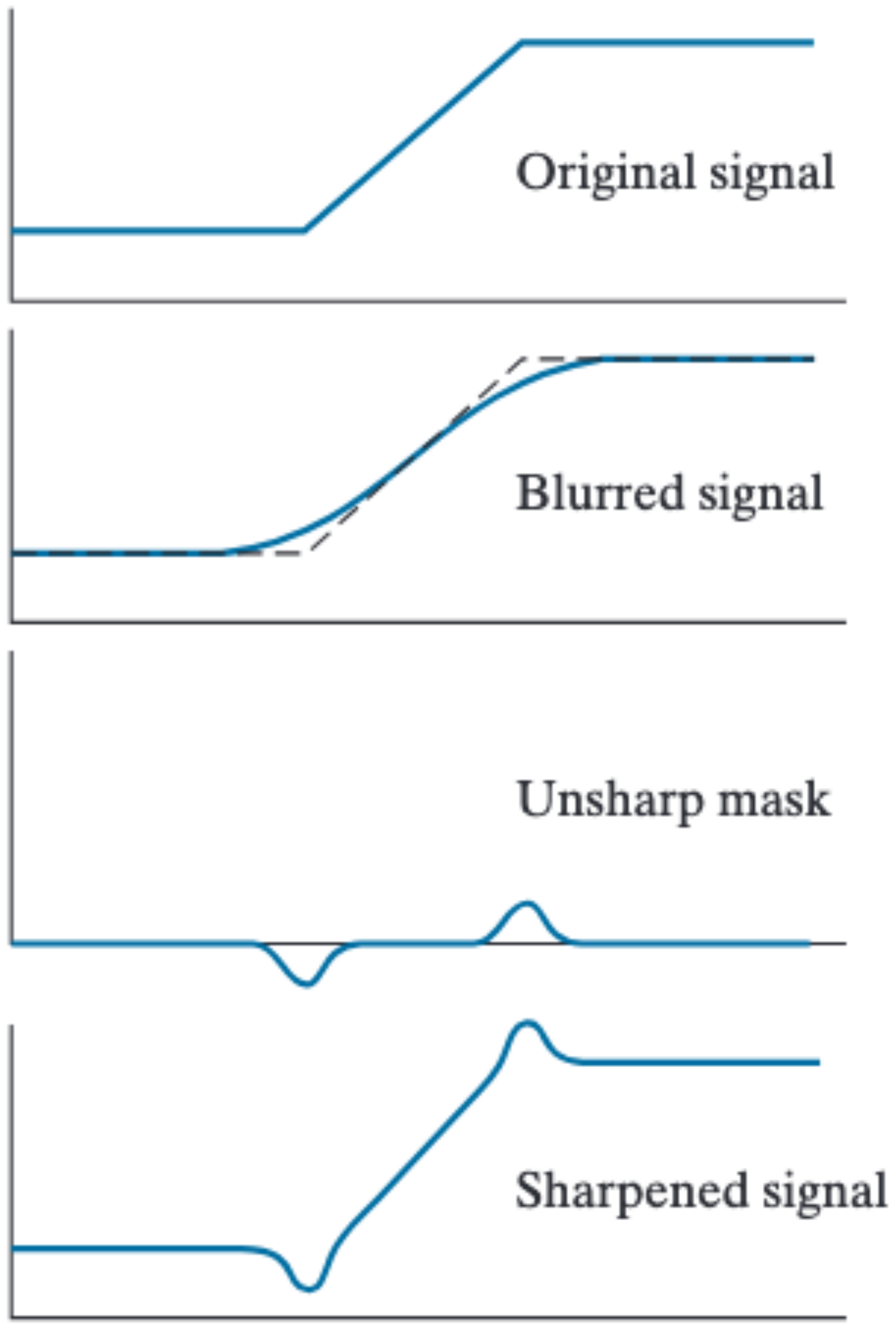
```
$ python demo.py
```

Filtro com Máscara de Unsharpening

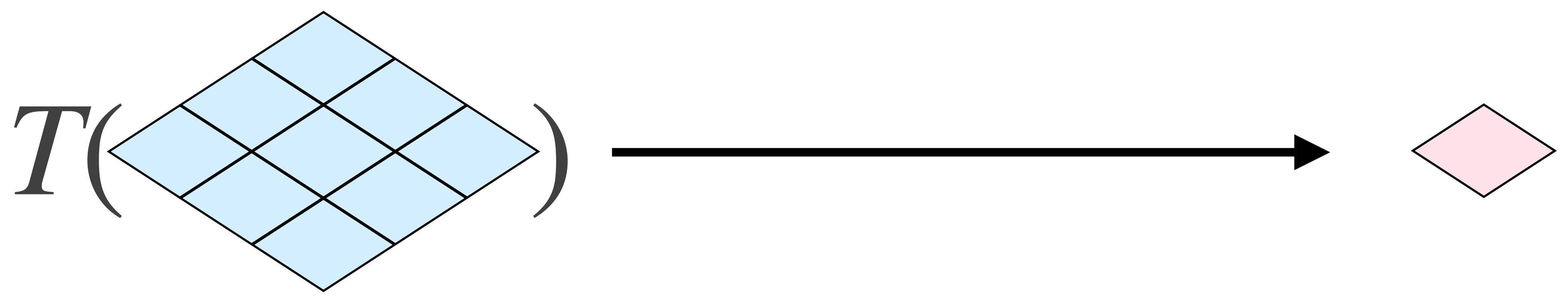
Ideia é usar diferença entre imagem suavizada e original para encontrar as bordas

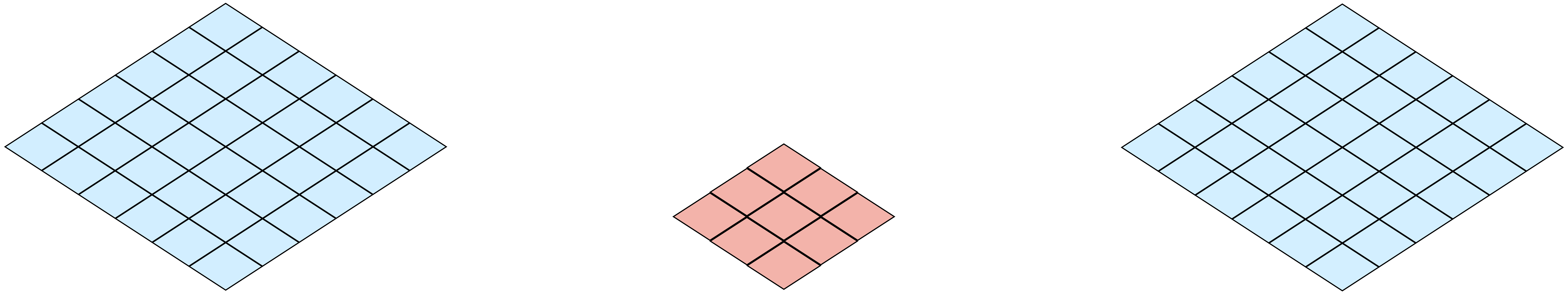
$$g_{mask} = f - \hat{f}$$

$$g(x, y) = f(x, y) + kg_{mask}(x, y)$$



Filtro de Gradiente





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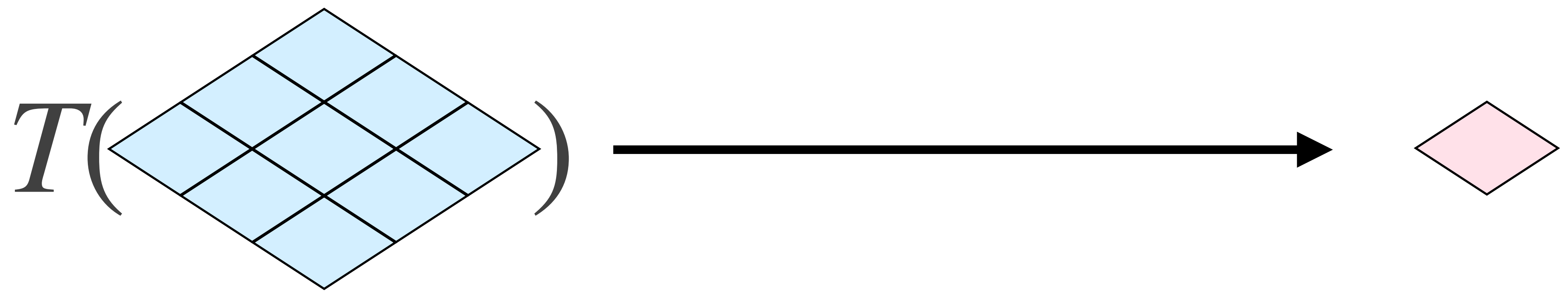
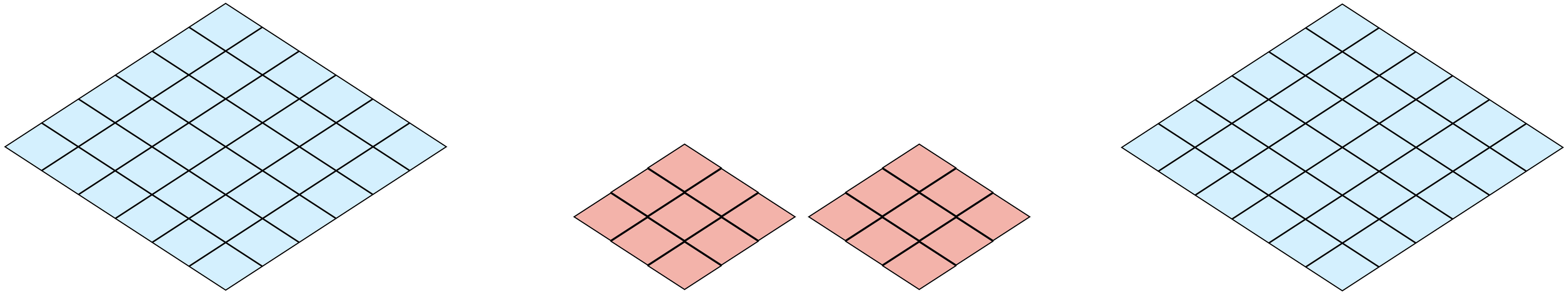
?	?	?

Filtro de Sobel



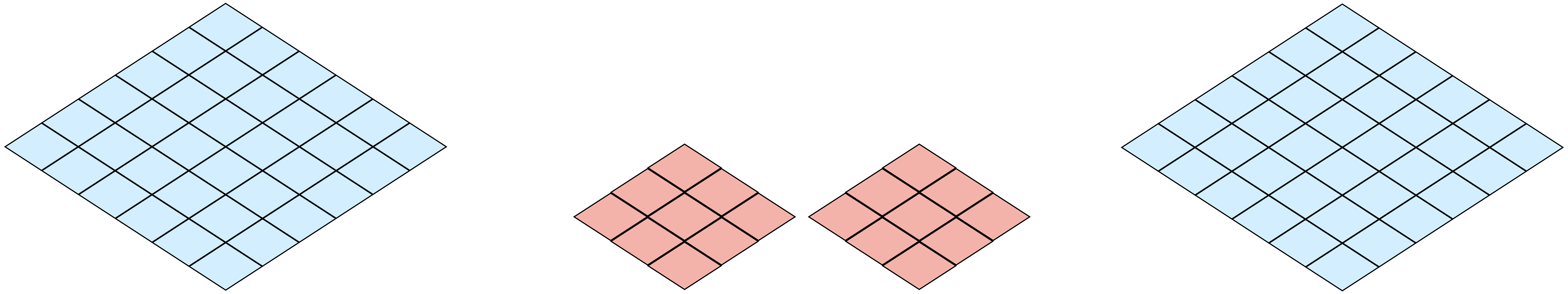


-1	-2	-1
0	0	0
1	2	1

-1	0	1
-2	0	2
-1	0	1

Filtro de Sobel

$$M(\nabla_f) = \sqrt{g_x^2 + g_y^2}$$



-1	-2	-1
0	0	0
1	2	1

-1	0	1
-2	0	2
-1	0	1

Filtros de Sobel

```
$ python demo.py
```

Processamento de Imagens

Filtros e Convolução (Parte II)

Professora Leo Sampaio Ferraz Ribeiro